SHARNA HOSSAIN

Software Engineer

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PROFESSIONAL SUMMARY

Software engineer experienced in modernizing complex legacy systems into scalable distributed architectures. Skilled in driving end-to-end system delivery, translating business workflows into production systems, and leveraging AI tools to accelerate development and improve engineering impact.

WORK EXPERIENCE

Software Engineer II

Morgan Stanley | August 2024 - Present | New York, NY

- Led end-to-end production delivery of a real-time payments modernization effort, overseeing recon, regression testing, and launch readiness for a critical settlement flow tied to a broader initiative projected to avoid $70M in future costs.
- Planned and drove modernization of a high-stakes monolithic system into a distributed architecture, using ADRs and C4 diagrams to define service boundaries and guide migration strategy.
- Renovated a legacy end-of-life GWT application with tightly coupled front-end and back-end components into a distributed web platform, modernizing the user interface and service layer to support long-term scalability and maintainability.
- Leveraged AI tooling to accelerate replication of legacy API behavior and support modernization efforts, improving development speed and reducing manual effort.
- Supported a high-priority firm initiative under a compressed timeline to enable real-time payments for a new trade model, ensuring daily FX trades in the millions executed reliably.

Software Engineer I

Morgan Stanley | February 2024 - August 2024 | New York, NY

- Architected and developed a microservice to replace a legacy Mainframe-dependent real-time settlements flow, modernizing a critical payment workflow and reducing reliance on a platform with approximately $200K in annual costs.
- Enabled real-time settlement processing and payment message flows between internal systems and agent banks, supporting journal creation and critical inter-entity payment operations.
- Authored architectural decision records and system design documentation, including C4 diagrams, to align stakeholders and drive implementation of a modernized settlement architecture.
- Delivered feature enablement to resolve inaccurate reference data ingestion, partnering with stakeholders and users to improve data quality and downstream system reliability.

EDUCATION

Bachelor of Arts, Computer Science | Columbia University | 2022

Associate in Science, Computer Science | Borough of Manhattan Community College | 2021

Coding Bootcamp Certificate | New York Code + Design Academy | 2019

SKILLS & TECHNOLOGIES

Languages: Java, SQL, Python, JavaScript

Frameworks & Systems: Spring Boot, Angular, RESTful APIs, Distributed Systems

Tools & Platforms: Apache Kafka, IBM MQ, Docker, Jenkins, Snowflake, Gradle, Linux, Git